Margaret Lowell

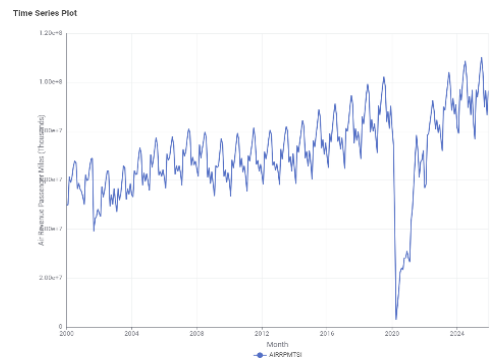
Forecast Project Memo

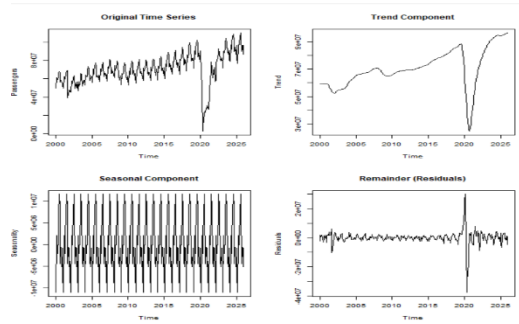
Overview and Data

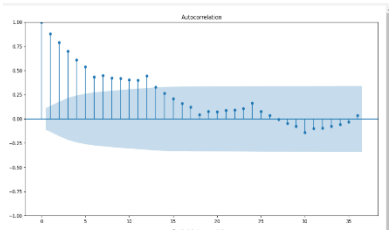
The objective of this project is to forecast air revenue passenger miles on United Staes domestic and international scheduled passenger flights. The dataset consists of monthly airline revenue (thousands) passenger miles (AIRRPMTSI) obtained from FRED. This data spans from January 2000 to December 2025. This data is collected by the U.S. Department of Transportation, Bureau of Transportation Statistics (BTS), U.S. Air Carrier Traffic Statistics reported in BTS' monthly Air Traffic Data. The goal of this forecast is to be able to plan for customer demand for flights. This is needed for capacity of passenger flights, staffing, scheduling decisions, pricing strategy, and potentially risk management.

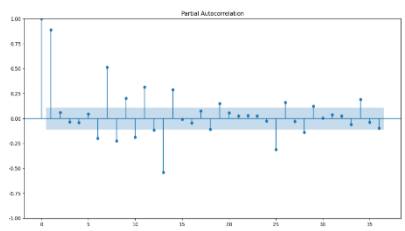
Initial Diagnostics

In the initial diagnostics, the original time series graph demonstrated an overall positive trend, clear seasonality, and an obvious disruption due to COVID-19. The ACF plot showed a gradual decrease suggesting that the data is not stationary. This suggested that the data needs differencing for ARIMA models. The ACF also suggests a moving average component of order 0 although this evidence isn't extremely clear through the plot. The PACF displays a significant spike at lag 1, indicating a potential autoregressive component of order 1. The ADF test resulted in a p-value of 0.24, which is greater than 0.05, also supporting the evidence that the data is not stationary. The Ljung-Box p-value of 0 suggests that patterns exist in the data.









#	RowID	adf_statistic Number (Float)	adf_pvalue Number (Float)	ljung_box_statistic Number (Float)	ljung_box_pvalue Number (Float)
1	1	-2.797	0.24	1,325.813	0

Model Building and Comparison

The baseline models used are seasonal naïve due to the seasonal component of the data and drift due to the strong positive trend in the data. For advanced models, ETS (A, A, A) was selected due to the consistent trend and the recurring seasonal patterns. The data didn't show an obvious increase in the seasonal swings through time. The trend and seasonal patterns were basically consistent over time suggesting that ETS (A, A, A) was a good candidate model. The other three advanced

models are three ARIMA models as suggested by the ACF and PACF plots. The first, ARIMA (1,1,0), is a candidate due to the diagnostic evidence explained above. ARIMA91,1,1) is also tested as the ACF evidence of a MA component order of 0 wasn't strong. Lastly, SARIMA (1,1,0) (1,0,0) [12], is tested to account for the yearly seasonality. After comparison of these six models, Time Series Cross Validation was used to determine which model provided better forecasts on my top performing models. The forecast horizon for the models is 12 periods considering there is strong seasonality.

Baseline and ETS(A,A,A) Model Diagnostics Table — Open in new window

Rows: 3 | Columns: 6

output_type (String)	RMSE (Number (Float))	MAE (Number (Float))	MAPE (Number (Float))	lb_pvalue (Number (Float))	Model (String)
diagnostics	1,408,294.572	1,081,131.333	1.177	0	Seasonal Naive
diagnostics	9,543,293.413	7,504,930	8.371	0	Drift
diagnostics	4,711,497.303	3,598,081.544	4.128	0	ETS(A,A,A)

ARIMA Models Diagnostic Table — Open in new window

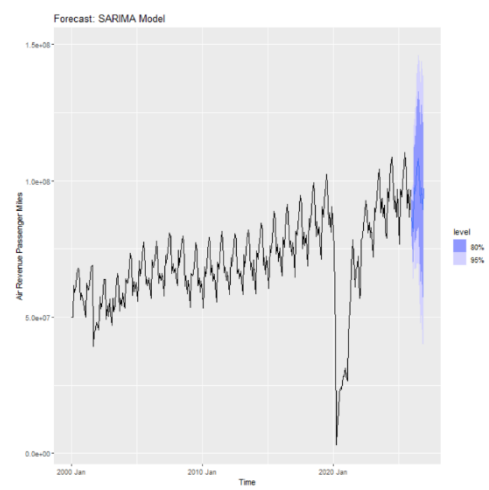
Rows: 3 | Columns: 5

sigma^2 (Number (Float))	log_lik (Number (Float))	AIC (Number (Float))	BIC (Number (Float))	Model (String)
28,120,168,334,243.617	-5,058.52	10,123.04	10,134.14	SARIMA (1,1,0) (1,0,0)[12]
35,415,107,286,132.73	-5,090.15	10,188.31	10,203.11	ARIMA(1,1,0)
35,536,160,853,917.09	-5,090.15	10,190.3	10,208.81	ARIMA(1,1,1)

Validation and Model Selection

Based on the diagnostics, Time-Series Cross Validation was performed on the Seasonal Naïve, ARIMA (1, 1, 0), and SARIMA (1,1,0) (1,0,0) [12] to see how accurate these models forecast. After TSCV, the best model is SARIMA (1,1,0) (1,0,0) [12]. This model had the lowest RMSE, MAE, MAPE, and ACF1 compared to the other two models. A MASE value of 0.59 suggests that the SARIMA model substantially outperforms the naive benchmark, reducing forecast error by 41%. Because of this, the SARIMA model is the selected model. However, the residual diagnostics indicate that some autocorrelation remains. This is seen in the high ACF1 value. The ACF1 values of all the models in the TSCV was about 0.8. This suggests that, while the models may not fully eliminate autocorrelation. The SARIMA model still captures the dominant patterns necessary for accurate prediction of demand. An explanation of this result could lie in the huge structural shock from the 2020 pandemic.

RowID	.model (String)	.type (String)	ME (Number (Float))	RMSE (Number (Float))	MAE (Number (Float))	MPE (Number (Float))	MAPE ↓ (Number (Float))	MASE (Number (Float))	RMSSE (Number (Float))	ACF1 (Number (Float))	Model (String)
Row1	SNAIVE(value ~ lag(	Test	1,501,883.466	18,232,185.929	8,454,698.992	-18.59	32.169	1.076	1.085	0.88	Seasonal Naive
Row0	ARIMA(value ~ pdq(	Test	-231,334.295	21,627,553.84	8,979,739.091	-17.376	31.895	1.143	1.287	0.886	ARIMA(1,1,0)
Row2	ARIMA(value ~ pdq(	Test	-2,956,995.373	6,717,525.469	4,639,889.172	-3.838	6.172	0.591	0.4	0.835	SARIMA

Forecast and Practical Implication

The forecast of the SARIMA model shows that airline passenger demand is expected to continue increasing over time, with regular seasonal peaks occurring during high-travel months. The practical implications of this are that airlines should increase flights during peak months, plan more staff during busy seasons, adjust pricing strategies (higher during peaks, promotions during lows), and prepare capacity ahead of increases in demand.